The HHL Algorithm

Quantum Linear Systems and Spectral Methods

Linear Systems Everywhere

- Solve:

$$A\mathbf{x} = \mathbf{b}$$

- Appears in:
  - PDEs
  - Machine learning
  - Many-body physics
- Classical cost: $\mathcal{O}(N^3)$ (dense case)

Quantum Goal

- Encode vector as quantum state:

$$|b\rangle = \sum_i b_i |i\rangle$$

- Goal:

$$|x\rangle \propto A^{-1}|b\rangle$$

- Output is a quantum state, not a classical vector

Eigenbasis Representation

$$A|u_j\rangle = \lambda_j|u_j\rangle$$

$$|b\rangle = \sum_j \beta_j|u_j\rangle$$

Formal Solution

$$A^{-1}|b\rangle = \sum_j \frac{\beta_j}{\lambda_j}|u_j\rangle$$

- Problem reduces to eigenvalue inversion

Hamiltonian Encoding

- Assume A is Hermitian
- Implement:

$$U = e^{iAt}$$

QPE Action

$$|u_j\rangle \to |u_j\rangle|\lambda_j\rangle$$

State After QPE

$$\sum_j \beta_j |u_j\rangle |\lambda_j\rangle$$

Key Step: Inversion

- Introduce ancilla qubit
- Perform controlled rotation:

$$|\lambda_j\rangle \rightarrow \sqrt{1 - \frac{C^2}{\lambda_j^2}}|0\rangle + \frac{C}{\lambda_j}|1\rangle$$

State After Rotation

$$\sum_j \beta_j |u_j\rangle |\lambda_j\rangle \left(\sqrt{1 - \frac{C^2}{\lambda_j^2}} |0\rangle + \frac{C}{\lambda_j} |1\rangle \right)$$

Inverse QPE

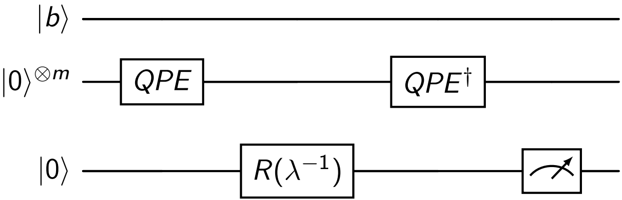
- Remove eigenvalue register
- Leaves:

$$\sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle$$

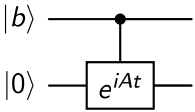
Final State

$$|x\rangle \propto A^{-1}|b\rangle$$

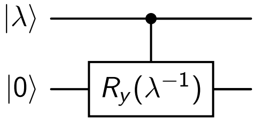
HHL Circuit (Overview)



Detailed Circuit Step 1: QPE



Detailed Circuit Step 2: Controlled Rotation



Detailed Circuit Step 3: Uncomputation

Post-selection

- Measure ancilla
- Condition on $|1\rangle$
- Success probability depends on κ

Runtime

$$\mathcal{O}\left(\kappa^2 \log N\right)$$

- $\kappa =$ condition number

Conditions for Speedup

- Sparse matrix
- Efficient Hamiltonian simulation
- Efficient state preparation

Readout Problem

- Cannot extract full vector efficiently
- Only expectation values accessible

Condition Number Issue

- Runtime scales with κ
- Ill-conditioned systems are hard

Operator View

$$A^{-1} = \sum_j \frac{1}{\lambda_j} |u_j\rangle\langle u_j|$$

Connection to Green's Functions

$$G = (A)^{-1}$$

- Resolvent operator
- Linear response theory

Many-Body Perspective

- HHL = spectral filtering
- Similar to:
  - Imaginary time evolution
  - Projection methods

Applications

- Quantum machine learning
- PDE solvers
- Data fitting

Summary

- HHL solves $Ax = b$ in quantum form
- Uses:
  - Phase estimation
  - Controlled rotation
  - Uncomputation
- Connects strongly to spectral physics

From Linear Systems to Resolvents

A central observation is that HHL solves a linear system

$$Ax = b$$

by preparing a quantum state proportional to

$$|x\rangle \propto A^{-1}|b\rangle.$$

In many-body physics, the inverse of an operator is precisely the basic structure of a *resolvent* or *Green's function*.
If we define

$$G(z) = (zI - H)^{-1},$$

then solving

$$(zI - H)|x\rangle = |b\rangle$$

is equivalent to preparing

$$|x\rangle = G(z)|b\rangle.$$

Thus HHL may be viewed as a quantum algorithm for applying a resolvent operator to a source state.

Spectral Form of the Green's Function

Let

$$H|u_j\rangle = E_j|u_j\rangle.$$

Then the resolvent has the spectral decomposition

$$G(z) = (zI - H)^{-1} = \sum_j \frac{1}{z - E_j}|u_j\rangle\langle u_j|.$$

If

$$|b\rangle = \sum_j \beta_j|u_j\rangle,$$

then

$$G(z)|b\rangle = \sum_j \frac{\beta_j}{z - E_j}|u_j\rangle.$$

This is structurally identical to the HHL transformation:

$$A^{-1}|b\rangle = \sum_j \frac{\beta_j}{\lambda_j}|u_j\rangle.$$

Hence, HHL implements the same spectral inversion mechanism that

Physical Meaning of the Spectral Filter

The factor

$$\frac{1}{z - E_j}$$

acts as a spectral filter.

- Eigenmodes with energies near z are enhanced.
- Eigenmodes far from z are suppressed.
- The inverse operator encodes propagation and response properties.

In this sense, HHL is not merely a linear solver. It is a *spectral transformation algorithm*: it maps a source state to a response state through an inverse operator.

Retarded Green's Function

In condensed matter and many-body theory, one often uses

$$G^R(\omega) = \frac{1}{\omega + i\eta - H}, \qquad \eta > 0.$$

The small positive η regularizes the inverse and enforces causal boundary conditions.
Applying this operator to a source state gives

$$|x(\omega)\rangle = G^R(\omega)|b\rangle.$$

Formally, an HHL-type algorithm can be interpreted as a way of preparing a quantum state proportional to this frequency-dependent response vector, provided one can encode the shifted operator

$$A(\omega) = \omega I - H$$

or

$$A(\omega) = \omega I + i\eta - H.$$

Connection to Correlation Functions

Many observables in quantum many-body physics can be written in terms of Green's functions or resolvents.
For an operator $\hat{O}$, a typical linear-response quantity has the form

$$\chi(\omega) = \langle \psi_0 | \hat{O}^\dagger \frac{1}{\omega + i\eta - (H - E_0)} \hat{O} | \psi_0 \rangle.$$

If we define the source state

$$|b\rangle = \hat{O} |\psi_0\rangle,$$

then

$$\chi(\omega) = \langle b | \frac{1}{\omega + i\eta - (H - E_0)} | b \rangle.$$

This shows that HHL-like inversion is directly connected to computing dynamical susceptibilities.

HHL as a Green's Function Engine

At a conceptual level, HHL performs three tasks:

1. Decompose the source state into eigenmodes.
2. Invert the spectral coefficients:

$$\lambda_j \mapsto \frac{1}{\lambda_j}.$$

3. Recombine the filtered eigenmodes into the response state.

This is exactly what is needed for:

- resolvent methods,
- Green's function methods,
- frequency-domain linear response,
- projection-based many-body calculations.

Linear Response in Physics

Suppose a system with Hamiltonian H_0 is perturbed by

$$V(t) = -f(t)\hat{F}.$$

To first order, the expectation value of an observable $\hat{O}$ changes as

$$\delta\langle \hat{O}(t)\rangle = \int dt'\, \chi_{OF}(t-t')f(t'),$$

where χ_{OF} is the response function.
In the frequency domain, one typically writes

$$\delta\langle \hat{O}(\omega)\rangle = \chi_{OF}(\omega)f(\omega).$$

The response function is often expressible through inverse operators of the form

$$(\omega I - M)^{-1},$$

which is precisely the structure targeted by HHL.

Resolvent Form of the Susceptibility

A standard spectral representation is

$$\chi_{OF}(\omega) = \sum_{n \neq 0} \left(\frac{\langle 0|\hat{O}|n\rangle\langle n|\hat{F}|0\rangle}{\omega - (E_n - E_0) + i\eta} - \frac{\langle 0|\hat{F}|n\rangle\langle n|\hat{O}|0\rangle}{\omega + (E_n - E_0) + i\eta} \right).$$

This is a sum over poles, or equivalently a matrix element of a resolvent. Thus linear response is fundamentally an *inverse-operator problem*, which places it in the same mathematical class as HHL.

What is Time-Dependent Hartree–Fock?

Time-dependent Hartree–Fock (TDHF) describes the evolution of a Slater determinant under a time-dependent mean field.
At the one-body density-matrix level,

$$i\hbar \frac{d\rho}{dt} = [h[\rho], \rho].$$

To study small oscillations around a stationary solution ρ_0, one linearizes:

$$\rho(t) = \rho_0 + \delta\rho(t),$$

leading to an equation of motion for the fluctuation $\delta\rho$.

Linearized TDHF

After linearization, one obtains a matrix equation of the form

$$i\hbar \frac{d}{dt}\delta\rho = \mathcal{L}\,\delta\rho + s(t),$$

where:

- $\mathcal{L}$ is the linearized Liouvillian or RPA matrix,
- $s(t)$ is an external driving term.

In the frequency domain, this becomes

$$(\omega I - \mathcal{L})\delta\rho(\omega) = s(\omega).$$

This is now explicitly a linear system. Its solution is

$$\delta\rho(\omega) = (\omega I - \mathcal{L})^{-1} s(\omega).$$

That is exactly the algebraic structure addressed by HHL.

TDHF as an HHL-Type Problem

The formal similarity is immediate:

$$Ax = b \qquad \Longleftrightarrow \qquad (\omega I - \mathcal{L})\delta\rho(\omega) = s(\omega).$$

Hence:

$$A \leftrightarrow \omega I - \mathcal{L}, \qquad x \leftrightarrow \delta\rho(\omega), \qquad b \leftrightarrow s(\omega).$$

If one can encode the linear-response matrix $\mathcal{L}$ into a quantum representation, then an HHL-type algorithm could in principle prepare a state proportional to the TDHF response.

Connection to Random Phase Approximation (RPA)

The small-amplitude limit of TDHF is closely related to the Random Phase Approximation (RPA).
One typically obtains a non-Hermitian eigenvalue problem of the form

$$\begin{pmatrix} A & B \\ -B^* & -A^* \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = \omega \begin{pmatrix} X \\ Y \end{pmatrix}.$$

Equivalently, one can formulate a driven linear-response problem

$$(\omega I - M)\, v = s.$$

Again, this is an inverse problem. The response amplitudes are obtained by applying the resolvent

$$(\omega I - M)^{-1}$$

to a source vector.

What HHL Would Compute in TDHF

In a TDHF or RPA context, one is often interested in:

- transition amplitudes,
- density oscillations,
- response to an external field,
- strength functions.

These quantities can often be expressed as matrix elements involving the response vector:

$$S(\omega) \sim \langle s|(\omega I - M)^{-1}|s\rangle.$$

Thus an HHL-type routine would not need to output the full vector $\delta\rho(\omega)$. It would already be useful if it allowed efficient estimation of such expectation values.

Strength Functions and Dynamical Response

A common observable in TDHF and RPA is the strength function

$$S_F(\omega) = -\frac{1}{\pi}\,\mathrm{Im}\,\chi_F(\omega),$$

where

$$\chi_F(\omega) = \langle 0|\hat{F}^\dagger \frac{1}{\omega + i\eta - (H - E_0)} \hat{F}|0\rangle.$$

This is exactly a Green's function expression. Hence:

$$\text{TDHF/RPA response} \quad \longleftrightarrow \quad \text{resolvent evaluation} \quad \longleftrightarrow \quad \text{HHL-type inver}$$

Why This Connection is Interesting

The HHL framework is potentially relevant because:

- linear response is naturally formulated as a linear system,
- inverse operators define propagators and susceptibilities,
- many observables are expectation values rather than full vectors,
- this aligns well with quantum output models.

From a physics viewpoint, HHL is best interpreted not only as a solver of $Ax = b$, but as a quantum method for preparing *response states* and probing *spectral properties*.

Important Caveats

There are, however, important obstacles:

- The matrix A must be efficiently block-encoded or simulated.
- TDHF/RPA matrices may be non-Hermitian.
- Condition numbers may be large near resonances.
- State preparation and measurement may dominate the cost.

Therefore, the connection is mathematically deep and physically appealing, but practical advantage depends strongly on encoding, sparsity, conditioning, and the observables of interest.

Hermitian Embedding of Non-Hermitian Problems

If the response matrix M is non-Hermitian, one common strategy is Hermitian embedding:

$$\mathcal{H} = \begin{pmatrix} 0 & M \\ M^\dagger & 0 \end{pmatrix}.$$

Then $\mathcal{H}$ is Hermitian, and one may reformulate the problem in an enlarged space.

This is often the natural bridge between:

- non-Hermitian response equations,
- quantum phase estimation or HHL-type methods,
- Green's function formulations on quantum hardware.

Conceptual Summary

Key idea

HHL applies an inverse operator spectrally:

$$A^{-1} = \sum_j \frac{1}{\lambda_j} |u_j\rangle\langle u_j|.$$

Green's function viewpoint

Choose

$$A = zI - H,$$

then HHL prepares a state proportional to

$$(zI - H)^{-1}|b\rangle.$$

TDHF / linear-response viewpoint

Choose

$$A = \omega I - \mathcal{L}$$